

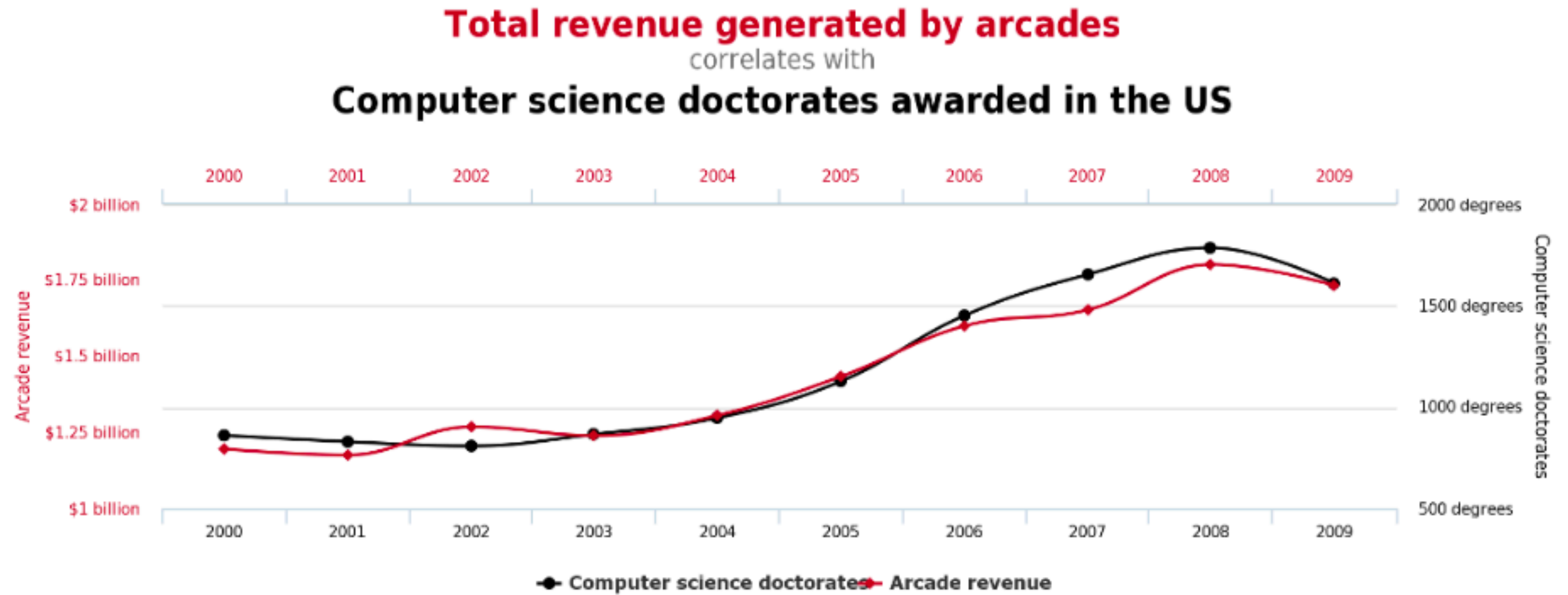
READINGS ON CONFIDENCE INTERVALS, HYPOTHESIS TESTING

Wasserman: 6.3, 10.2, 10.3 , 10.7

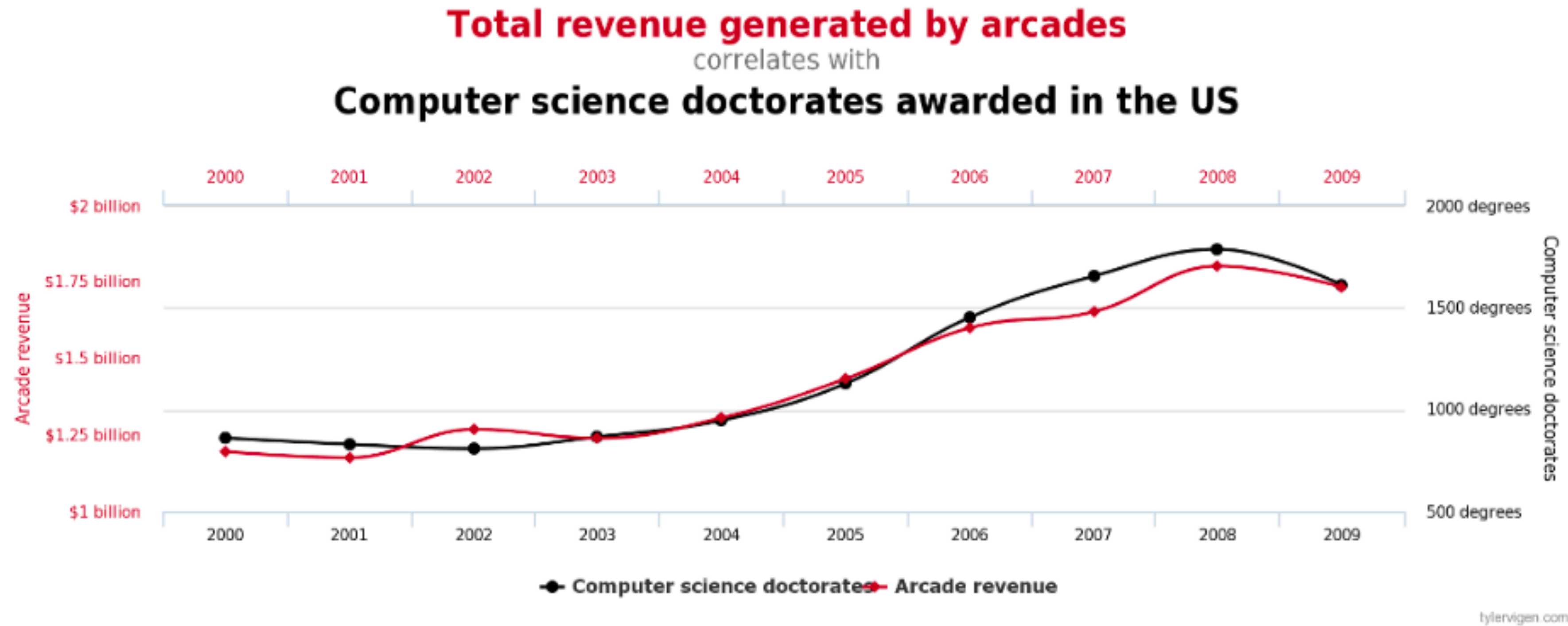
Watkins: 16.1, 16.2, 17.1, 18.1, 18.2, 18.3, 18.5, 20.1, 20.2, 20.3,
20.5, 20.6, 21, 2

One last topic before we end our discussion on hypothesis testing!

MULTIPLE TESTING CORRECTION



MULTIPLE TESTING CORRECTION



Thousands of hypotheses tested (correlations of pairs of variables).

Significance level of 0.05 not a high standard in this case:

Buying a single lottery ticket and winning vs
Buying millions of tickets and winning.

MULTIPLE TESTING CORRECTION

Bonferroni Correction:

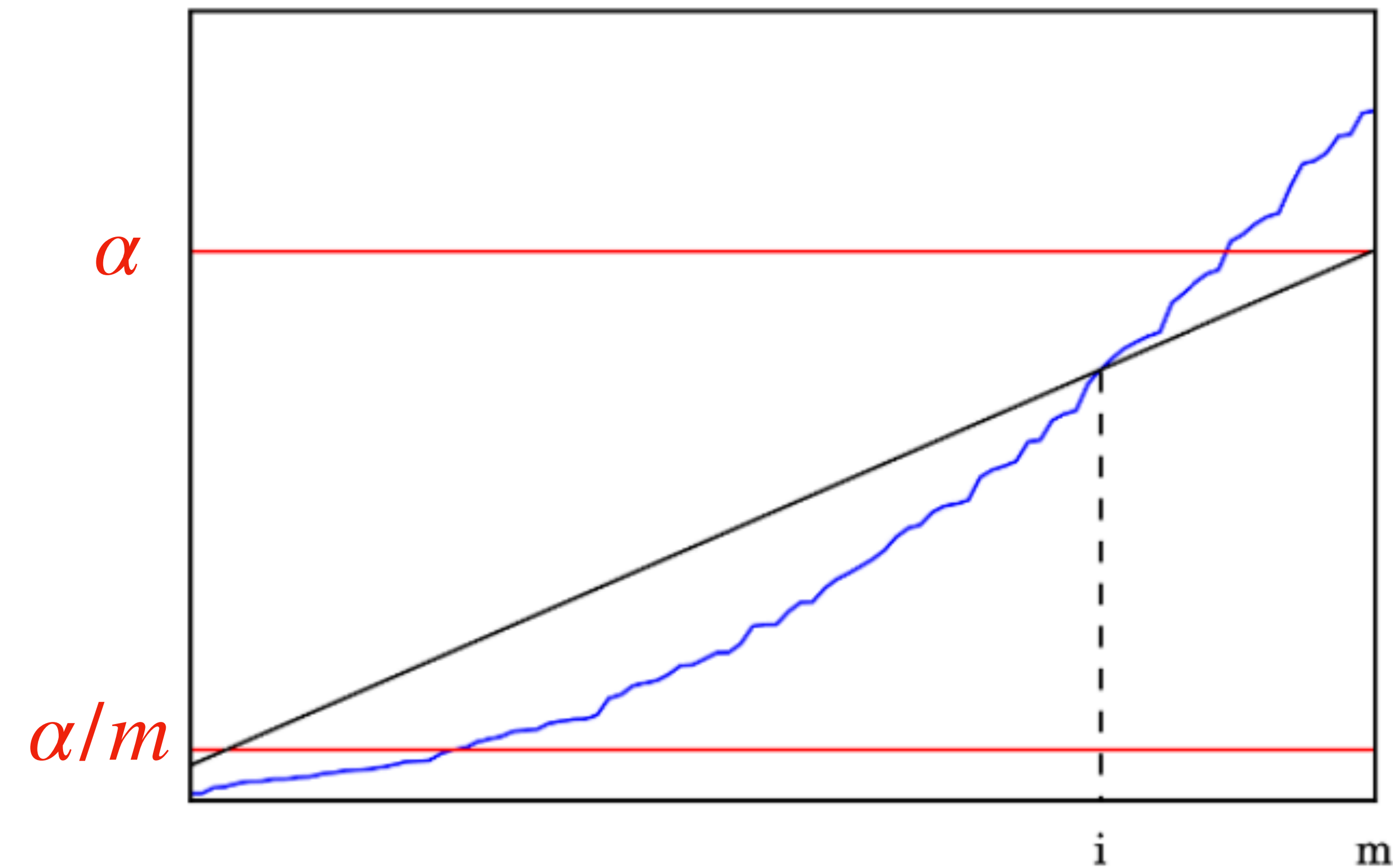
If testing m hypothesis $\Rightarrow$ to be significant at α , p-value against α/m .

MULTIPLE TESTING CORRECTION

Bonferroni Correction:

If testing m hypothesis $\Rightarrow$ to be significant at α , p-value against α/m .

Benjamini Hochberg Procedure:



p_i : p-value of the i^{th} hypothesis

Sort all p_i in ascending order.

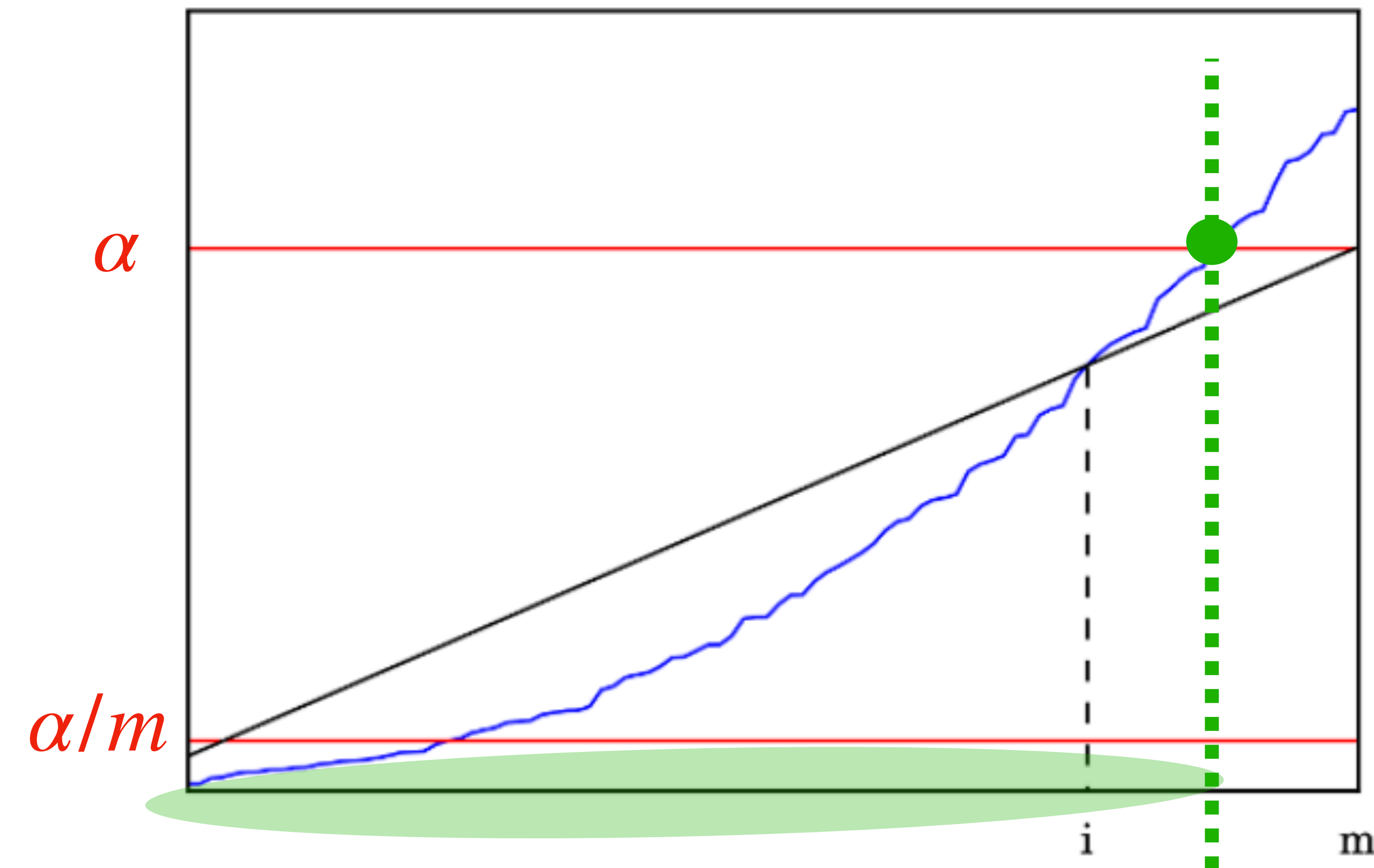
Blue curve: p-values

MULTIPLE TESTING CORRECTION

Bonferroni Correction:

If testing m hypothesis $\Rightarrow$ to be significant at α , p-value against α/m .

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p_i : p-value of the i^{th} hypothesis

Sort all p_i in ascending order.

Blue curve: p-values

If check against α , each j in shaded region is significant.

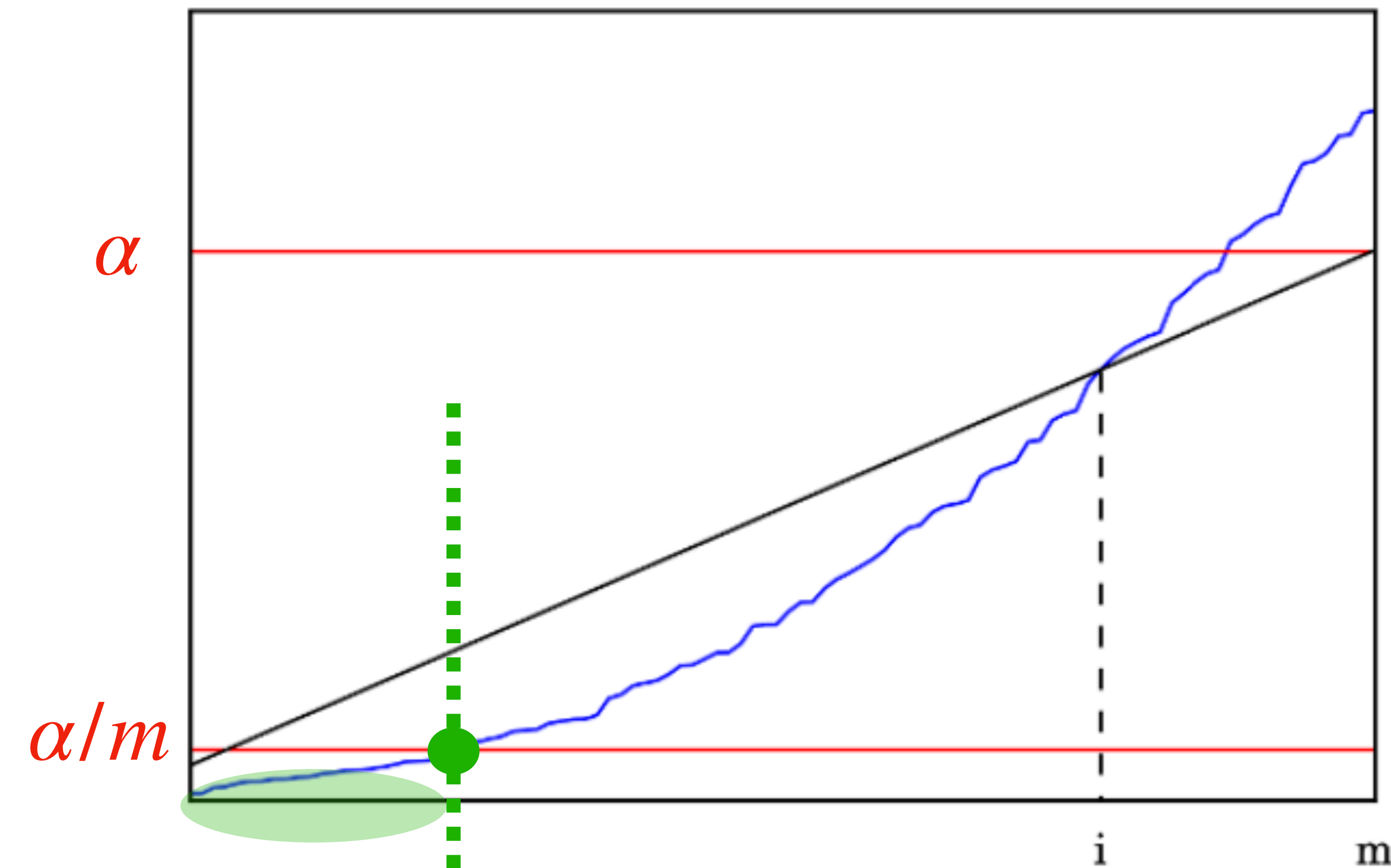
No correction! Too relaxed.

MULTIPLE TESTING CORRECTION

Bonferroni Correction:

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Benjamini Hochberg Procedure:



p_i : p-value of the i^{th} hypothesis

Sort all p_i in ascending order.

Blue curve: p-values

If check against α/m , each j in shaded region is significant.

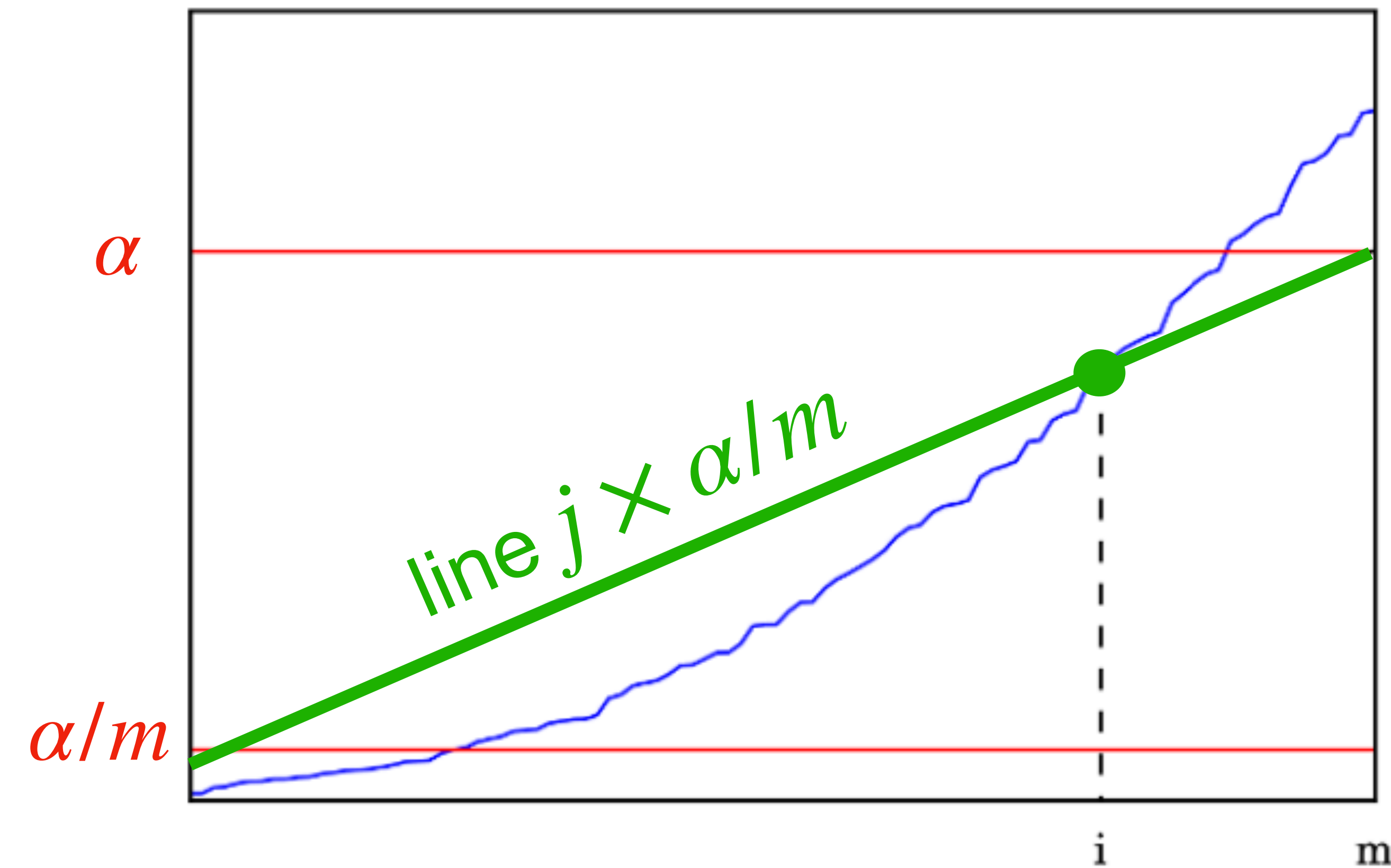
Bonferroni correction! Too strict.

MULTIPLE TESTING CORRECTION

Bonferroni Correction:

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Benjamini Hochberg Procedure:



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Sort all p_i in ascending order.

Blue curve: p-values

Let j represent x-coord.

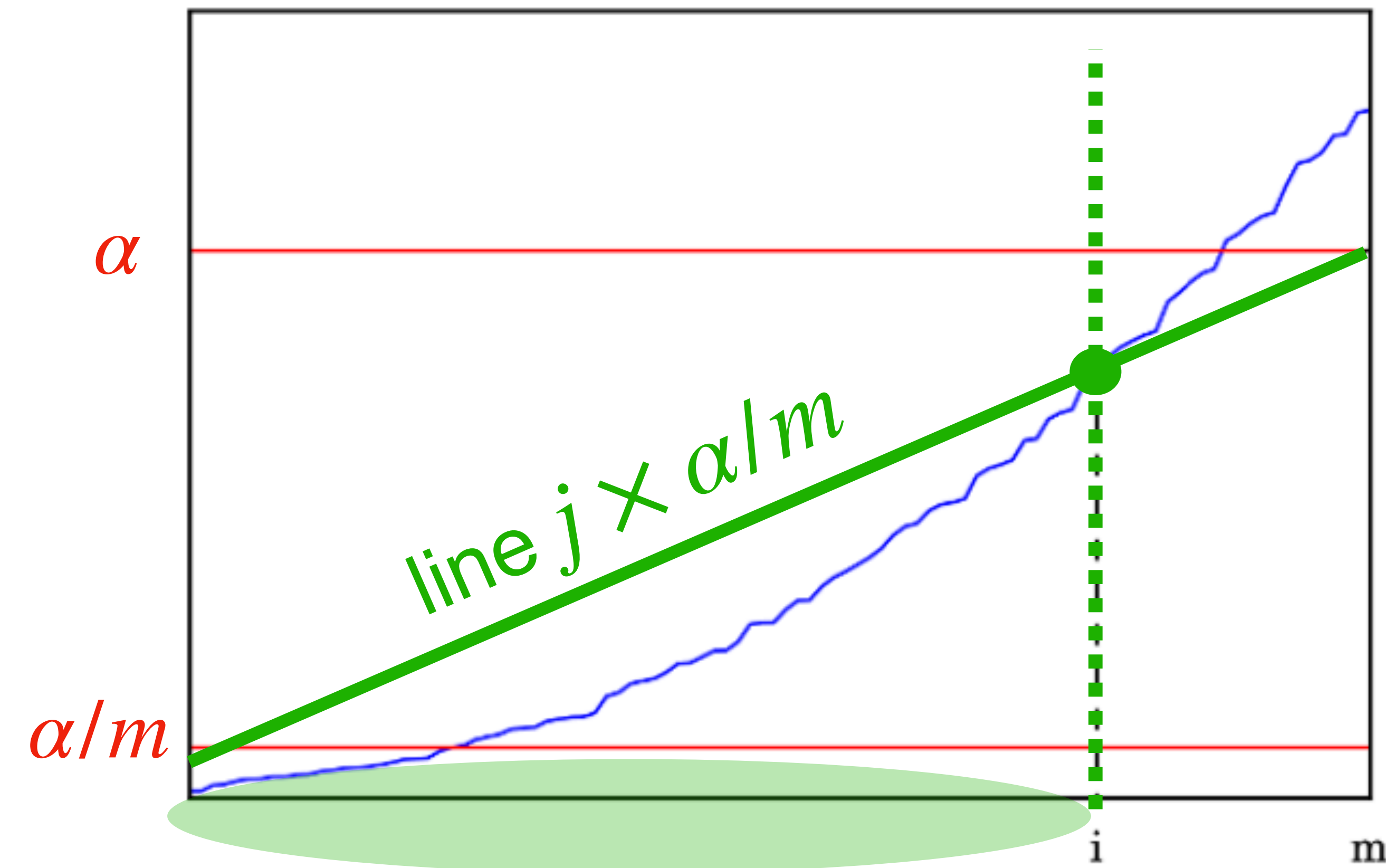
Consider line $j \times \alpha/m$.

MULTIPLE TESTING CORRECTION

Bonferroni Correction:

If testing m hypothesis $\Rightarrow$ to be significant at α , p-value against α/m .

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Let j represent x-coord.

Consider line $j \times \alpha/m$.

Variables up to i , where

$\forall_{j=1}^i p_j \leq j \times \alpha/m$ are significant.

PRELIMINARIES FOR EDA IN PYTHON

A notebook for a quick review of NumPy and Pandas at:

https://colab.research.google.com/drive/17Zx64-WzB6Mokvy3CN_fsQmyGCp7dj_C?usp=sharing

For further practice on NumPy:

Chapter 4 from Python for Data Analysis by Wes McKinney

<https://wesmckinney.com/book/numpy-basics>

For further practice on Pandas:

Chapter 5 from Python for Data Analysis by Wes McKinney

<https://wesmckinney.com/book/pandas-basics>

Chapter 6 from Learning Data Science by Sam Lau et al.

https://learningds.org/ch/06/pandas_intro.html

MORE ON DATAFRAMES

A notebook for an exploratory analysis of TED talks data at:

<https://colab.research.google.com/drive/1Ohqj99tdIPNkZ6fsvrKs6jjVMJ335S8X?usp=sharing>